

RARAU, Romania

WMO: 150520

Lat: 47.4496N

Lon: 25.5680E

Elev: 1574

StdP: 83.79

Time Zone: 2.00 (EUE)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.2	-16.1	-23.5	0.5	-11.9	-19.2	0.8	-12.8	19.6	-5.4	14.4	-6.1	3.5	0	0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	20.7	14.5	18.9	14.2	17.3	13.4	15.9	18.7	14.9	17.7	14.1	16.5	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	12.8	16.9	14.0	12.1	15.9	13.1	11.4	15.1	50.5	18.8	47.5	17.7	44.9	16.5	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.7	10.5	9.7														
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-21.3	18.6	3.8	2.0	-24.1	20.0	-26.3	21.2	-28.5	22.3	-31.2	23.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-6.0	-5.9	-4.1	1.1	6.8	10.4	12.0	12.0	7.1	3.8	-2.0	-5.9
	DBStd	8.24	5.08	5.27	5.31	4.37	4.16	3.53	3.22	3.68	4.43	5.19	5.48	4.42
	HDD10.0	2994	495	446	437	267	113	38	17	21	110	199	359	493
	HDD18.3	5782	753	679	695	517	356	239	196	197	337	451	609	751
	CDD10.0	257	0	0	0	0	15	49	80	83	23	6	0	0
	CDD18.3	2	0	0	0	0	0	0	1	1	1	0	0	0
	CDH23.3	6	0	0	0	0	0	1	2	2	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	0	1	0	0	0
(14) Wind	WSAvg	2.8	3.0	3.5	3.6	2.9	2.7	2.2	2.1	1.9	2.7	2.7	3.1	3.4
(15) Precipitation	PrecAvg	732	26	30	39	56	91	113	114	83	67	52	35	35
	PrecMax	1099	46	60	74	109	167	232	230	127	176	126	71	66
	PrecMin	505	6	8	10	23	43	60	50	46	22	6	0	10
	PrecStd	126	11	11	17	22	37	42	51	24	32	29	19	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	6.7	10.6	12.2	18.7	21.2	22.0	23.4	20.9	16.5	11.0	6.0
		MCWB	-0.4	1.7	5.3	8.0	11.4	15.2	15.1	16.1	13.5	11.6	6.3	2.5
	2%	DB	3.7	4.1	7.3	10.0	15.8	18.6	20.1	21.0	17.2	13.6	8.7	2.1
		MCWB	-0.6	0.9	3.3	6.6	10.3	14.4	15.0	14.7	12.9	9.5	5.3	0.5
	5%	DB	1.9	2.5	5.0	8.5	14.1	16.9	18.2	18.9	15.3	12.1	7.1	1.0
		MCWB	-0.8	0.1	2.4	5.7	10.0	12.8	14.0	14.4	11.8	8.8	5.0	0.4
	10%	DB	0.5	0.9	3.2	6.8	12.5	15.3	16.9	17.0	13.1	10.5	5.5	-0.2
		MCWB	-0.9	-0.7	1.4	4.6	9.2	12.3	13.6	13.6	10.8	8.2	3.7	-1.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.7	5.8	9.2	12.9	16.3	17.7	17.5	15.2	12.2	8.5	2.7
		MCDB	3.2	4.3	8.5	11.2	15.1	19.9	19.7	20.5	17.8	14.4	9.0	4.0
	2%	WB	0.9	1.6	3.9	7.2	11.4	15.0	15.7	16.0	13.5	10.7	6.4	1.5
		MCDB	1.7	3.3	6.1	9.0	14.4	17.7	17.9	19.1	16.2	12.4	7.6	2.2
	5%	WB	0.0	0.5	2.6	5.9	10.5	13.8	14.8	14.9	12.2	9.8	5.1	0.5
		MCDB	0.9	2.0	4.4	8.1	13.2	16.0	17.2	17.9	14.7	11.3	6.3	0.8
	10%	WB	-0.9	-0.5	1.5	4.8	9.6	12.8	13.9	14.0	11.1	8.5	4.0	-0.7
		MCDB	0.1	0.6	3.3	6.5	11.9	14.7	16.3	16.3	13.0	10.0	5.4	-0.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.8	6.0	5.7	6.4	6.4	6.4	6.3	6.0	6.3	5.2	5.0
		MCDBR	5.6	7.3	8.6	7.5	8.8	8.5	8.9	8.6	8.5	8.0	6.0	5.6
		MCWBR	4.2	5.4	5.5	4.7	5.0	5.1	5.0	4.7	5.4	5.3	4.6	4.6
	5% WB	MCDBR	5.1	6.6	7.8	6.8	7.2	8.0	7.8	7.5	7.6	6.7	5.5	5.1
		MCWBR	4.3	5.2	5.5	4.7	4.8	5.4	5.2	4.9	5.6	5.2	4.6	4.5
(40) Clear-Sky Solar Irradiance	taub		0.253	0.277	0.310	0.357	0.342	0.340	0.359	0.363	0.329	0.301	0.278	0.255
	taud		2.426	2.386	2.382	2.316	2.408	2.428	2.371	2.360	2.438	2.513	2.526	2.475
	Ebn at Noon		853	899	914	896	921	922	898	879	883	861	818	810
	Edh at Noon		65	85	100	119	113	111	116	112	94	75	60	56
(44) All-Sky Solar Radiation	RadAvg		1.20	2.02	3.11	4.17	5.13	5.45	5.43	5.01	3.58	2.42	1.35	0.93
	RadStd		0.18	0.24	0.40	0.42	0.52	0.45	0.51	0.46	0.46	0.33	0.19	0.13

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.71	N/A	N/A	+0.95	N/A	N/A	-157	-220	+103	+35	

Nomenclature: See separate page